```
!pip install matplotlib seaborn pandas numpy
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (
```

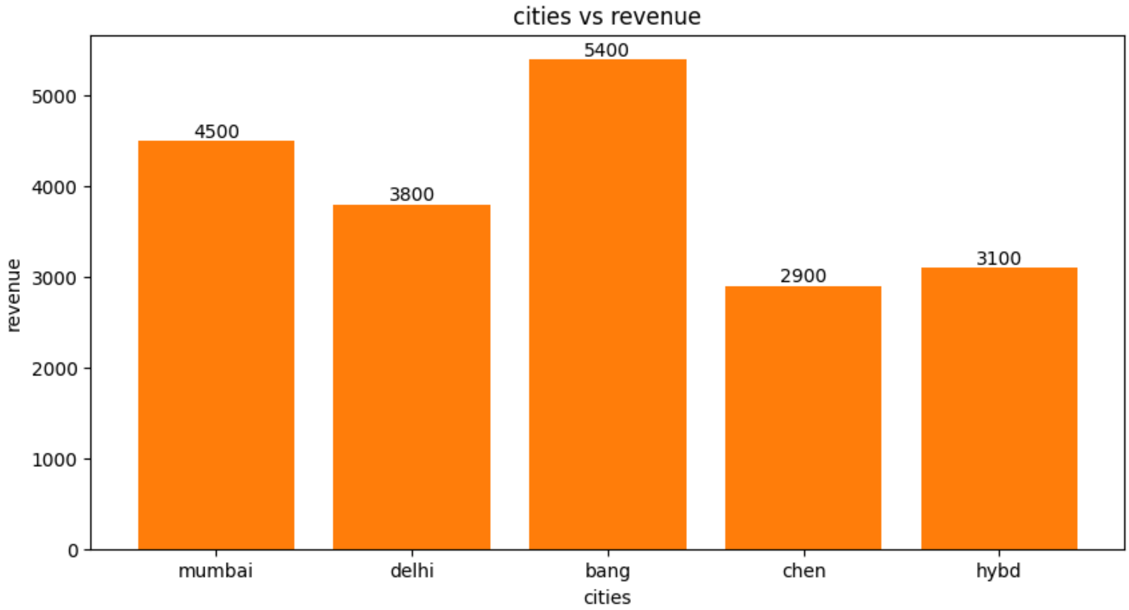
Start coding or generate with AI.

```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

```
age=[1,2,3,4,5,6,7]
salary=[100,200,300,400,500,600,700]
id=[1,2,3,4,5,6,7]
plt.figure(figsize=(10, 6))
plt.plot(age,salary,marker='o',linestyle='--',color='r')
plt.plot(age,id,marker="^",label="id")
plt.legend()
plt.title('Age vs salary')
plt.xlabel('Age')
plt.ylabel('salary')
plt.grid(True,alpha=0.3)
plt.savefig('age_vs_salary.png')
plt.show()
```


```
cities=['mumbai','delhi','bang','chen','hybd']
revenue=[4500,3800,5400,2900,3100]
fig,ax=plt.subplots(figsize=(10,5))
bars=ax.bar(cities,revenue)
ax.bar(cities,revenue)
for bar in bars:
  height=bar.get_height()
```

```
  ax.text(bar.get_x()+bar.get_width()/2,height,str(height),ha='center',va='bottom')
ax.set_xlabel('cities')
ax.set_ylabel('revenue')
ax.set_title('cities vs revenue')
plt.show()
```




```
sns.set_theme(style="darkgrid")
df=sns.load_dataset('titanic')
print(df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

Start coding or generate with AI.

```
# IMPORT LIBRARIES
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style='whitegrid')

# LOAD DATASET
data = {
    "Athlete": ["Usain Bolt", "Michael Phelps", "Simone Biles", "Neeraj Chopra", "Allyson Felix",
                "Carl Lewis", "Katie Ledecky", "Abhinav Bindra", "Mo Farah", "Shelly-Ann Fraser"],

    "Country": ["Jamaica", "USA", "USA", "India", "USA",
                "USA", "USA", "India", "UK", "Jamaica"],

    "Sport": ["Athletics", "Swimming", "Gymnastics", "Athletics", "Athletics",
              "Athletics", "Swimming", "Shooting", "Athletics", "Athletics"],

    "Gold": [8, 23, 7, 1, 7, 9, 7, 1, 4, 3],
    "Silver": [0, 3, 2, 0, 3, 1, 3, 0, 0, 4],
    "Bronze": [0, 2, 2, 0, None, 0, 1, 0, 0, 0]
}

df = pd.DataFrame(data)

print(df.shape)
print(df.dtypes)
print(df.head(3))
```

```
# STEP 1: IDENTIFY MISSING VALUES
print(df.isnull().sum())

# STEP 2: CLEAN DATA
df['Bronze'] = df['Bronze'].fillna(0)

# STEP 3: FIX DATA TYPES
df['Bronze'] = df['Bronze'].astype('int')

# STEP 4: VERIFY CLEANING
print('Missing after cleaning:', df.isnull().sum().sum())
print(df.describe())

# FEATURE ENGINEERING
df['Total'] = df[['Gold', 'Silver', 'Bronze']].sum(axis=1)
# Q1: Who is the top athlete?
top_athlete = df.sort_values(by='Total', ascending=False).iloc[0]
print("\nTop Athlete:\n", top_athlete)

# Q2: Total medals by country
country_total = df.groupby('Country', observed=False)['Total'].sum()
print("\nTotal Medals by Country:\n", country_total)

# Q3: Count of athletes by sport
sport_count = df.groupby('Sport', observed=False)['Sport'].count()
print("\nSport Counts:\n", sport_count)

# Q4: Average gold medals
avg_gold = df['Gold'].mean()
print(f"\nAverage Gold Medals: {avg_gold:.2f}")

# Q5: Total medals won by India
india_total = df[df['Country'] == 'India']['Total'].sum()
print("\nIndia Total Medals:", india_total)
```

```
(10, 6)
Athlete     object
Country     object
Sport       object
Gold         int64
Silver       int64
Bronze     float64
dtype: object
          Athlete  Country       Sport  Gold  Silver  Bronze
0      Usain Bolt  Jamaica   Athletics     8       0     0.0
1  Michael Phelps      USA    Swimming    23       3     2.0
2    Simone Biles      USA  Gymnastics     7       2     2.0
Athlete    0
Country    0
Sport      0
Gold       0
Silver     0
Bronze     1
dtype: int64
Missing after cleaning: 0
            Gold     Silver     Bronze
count  10.000000  10.000000  10.000000
mean    7.000000   1.600000   0.500000
std     6.306963   1.577621   0.849837
min     1.000000   0.000000   0.000000
25%     3.250000   0.000000   0.000000
50%     7.000000   1.500000   0.000000
75%     7.750000   3.000000   0.750000
max    23.000000   4.000000   2.000000

Top Athlete:
 Athlete    Michael Phelps
Country               USA
Sport            Swimming
Gold                   23
Silver                  3
Bronze                  2
Total                  28
Name: 1, dtype: object

Total Medals by Country:
 Country
India       2
Jamaica    15
UK          4
USA        70
Name: Total, dtype: int64

Sport Counts:
 Sport
Athletics     6
Gymnastics    1
```

```
Shooting      1
Swimming      2
Name: Sport, dtype: int64

Average Gold Medals: 7.00
```

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